

Properties and structure of $(1 - x)\text{Li}_2\text{O}-x\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-4\text{SiO}_2$ glass systems

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Abstract

$(1 - x)\text{Li}_2\text{O}-x\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-4\text{SiO}_2$ glasses were studied for the progressive percentage substitution of Na_2O for Li_2O at the constant mole of Al_2O_3 and SiO_2 . The crystallization temperature at the exothermic peak increased from 898 to 939 °C when the Na_2O content increases from 0 to 0.6 mol. The coefficient of thermal expansion and density of these as-quenched glasses increase from $6.54 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$ to $10.1 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$ and 2.378 g cm^{-3} to 2.533 g cm^{-3} when the Na_2O content increases from 0 to 0.4 mol, respectively. The electrical resistivity has a maximum value at $\text{Na}_2\text{O} \cdot (\text{Li}_2\text{O} + \text{Na}_2\text{O})^{-1} = 0.4$. The activation energy of crystallization decreases from 444 to 284 kJ mol^{-1} when the Na_2O content increased from 0 to 0.4 mol. Moreover, the activation energy increases from 284 kJ mol^{-1} to 446 kJ mol^{-1} when the Na_2O content increased from 0.4 to 0.6 mol. The FT-IR spectra show that the symmetric stretching mode of the SiO_4 tetrahedra ($1035\text{--}1054 \text{ cm}^{-1}$) and AlO_4 octahedra ($713\text{--}763 \text{ cm}^{-1}$) exhibiting that the network structure is built by SiO_4 tetrahedra and AlO_4 .

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1. Introduction

$\text{Li}_2\text{O} \cdot \text{Al}_2\text{O}_3 \cdot n\text{SiO}_2$ glass-ceramics (n is 2 for eucryptite and β -eucryptite, 4 for spodumene, 6 for lithium orthoclase, and 8 for petalite) are endowed with ultralow and even negative thermal expansion, high thermal stability, and high chemical durability [1]. They have been widely used as cook-top panels, stove windows, cookware, and some precision parts [2].

Conventional, low coefficient of thermal expansion ceramics such as β -spodumene, are fabricated mainly by the traditional recrystallization of solidified melt, i.e. preparation of monolithic glass products by the usual glass

molding technique including blowing, pressing or casting, followed by controlled crystallization.

β -Eucryptite ($\text{Li}_2\text{O} \cdot \text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2$) is structurally similar to β -quartz, the high temperature form of quartz which is unstable at ambient temperature. β -spodumene ($\text{Li}_2\text{O} \cdot \text{Al}_2\text{O}_3 \cdot 4\text{SiO}_2$) is derived from keatite, a form of silica which can only be prepared under hydrothermal conditions [3–5]. Both of these crystal phases are referred to as the forms of silica in which network replacement of Si^{4+} ions by Al^{3+} ions has occurred, accompanied by filling of interstitial vacancies by suitable cations, in these cases alkali ions, maintain electrical neutrality and stabilize the structures [6,7]. An knowledge of the change in properties with chemical composition is important to glass technologists. Since the glass composition must be capable of being melted and shaped by economical methods, the influence of composition must be considered in formulating glasses for low coefficient of thermal expansion materials. The pronounced

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change in properties resulting from the addition of a second alkali oxide to a glass has been called the mixed alkali effect [8]. The most prominent mixed alkali effect for oxide glasses is the pronounced reduction in direct current conductivity observed for alkali composition [8,9].

Although there have been a large number of studies concerning the properties and structure of R_2O – SiO_2 [10,11], R_2O – Al_2O_3 – SiO_2 [12–14], R_2O – CaO – Al_2O_3 – SiO_2 – TiO_2 [15,16], Li_2O – Al_2O_3 – SiO_2 – TiO_2 [17] glasses. An extensive literature search shows that details of the chemical resistance, electrical resistivity, optical properties and structure of mixed alkali calcium aluminogermanate glass systems have been studied so far [18–20]. However, the properties and structure related to the substitution of Na_2O to the Li_2O – Al_2O_3 – $4SiO_2$ glasses have not been elaborated.

2. Experimental procedures

2.1. Glass preparation

Reagent-grade of SiO_2 , Li_2CO_3 , Na_2CO_3 , and Al_2O_3 powders were used as raw materials for preparing the $(1-x)Li_2O$ – xNa_2O – Al_2O_3 – $4SiO_2$ glasses. The compositions of these glasses are listed in Table 1. Na_2O was added in 0.2 mol stepwise substitution for Li_2O in the $(1-x)Li_2O$ – xNa_2O – Al_2O_3 – $4SiO_2$ glasses except C_4 sample. All samples were obtained in 200 g batches by melting in a platinum crucible at 1500 °C in an electric furnace for 2 h. After homogenizing, the melt is cast onto a stainless steel plate and then annealed it at 450 °C for 2 h.

2.2. Samples characterization

Differential thermal analysis (DTA) on a 200 mg powder sample (67, 48 μm) in size at various heating rates from 5 to 20 °C min^{-1} in air was conducted with a Perkin Elmer DTA 7 Simultaneous Symmetrical Thermoanalyzer (USA) using powdered Al_2O_3 as a reference material.

A typical thermal expansion specimen was 10 mm long, 5 mm thick and 5 mm wide. Each sample was heated to the softening temperature in a dilatometer (Kirtier Scientific Instruments Co., Ltd, ROC) at a heating rate of 3 °C/min and the coefficient of thermal expansion was measured for the temperature range from 25 to 450 °C. The glass

transition and dilatometric softening temperatures were obtained from the thermal expansion curves.

Fourier transform infrared spectra were measured in the 400–1200 cm^{-1} region, using a Perkin Elmer Spectrum One FT-IR spectrometer (Boston, USA), with a resolution of 2 cm^{-1} . KBr pelletized disks were fabricated, containing 5 mg of the glass sample and 50 mg KBr.

The specimen for electrical resistivity measurement was about 10.0 $\times$ 5.0 $\times$ 5.0 mm. After grinding and polishing for parallel faces, the specimen was dashed coated with aluminum gel and dried at 105 °C for 24 h. The electrical resistivity was recorded with a HEWLETT 4156B resistance meter.

The density was measured with an Archimedes technique. The sample for density measurement was measured in air (W_{air}). Then the sample was boiled in water for 30 min to remove the air bubbles on the sample and measured in water (W_{water}). The density was calculated by Archimedes formula:

$$\rho \text{ (g cm}^{-3}\text{)} = (W_{air} \times D_{liq})(W_{air} - W_{liq})^{-1}, \quad (1)$$

where D_{liq} is the density of water, W_{air} is the weight measured in air, W_{liq} is the weight measured in water.

3. Results

3.1. Results of thermal analysis

The typical DTA curve obtained from the as-quenched glass C_1 at a heating of 5 °C min^{-1} is shown in Fig. 1. The first endothermic peak at 797 °C, corresponds to the glass transition point and the second one at 898 °C to the crystallization temperature (T_c) of the glass C_1 . In Fig. 1, the single glass transition temperature (T_g) suggests the existence of compositionally homogenous amorphous state in the as-quenched glass C_1 .

The typical DTA curves of the $(1-x)Li_2O$ – xNa_2O – Al_2O_3 – $4SiO_2$ ($0 \leq x \leq 1$) glass samples at a heating rate of 5 °C min^{-1} are shown in Fig. 2. It is found that the glass samples have the ubiquitously single T_g temperature. Moreover, all the glass samples also have one T_c except the C_6 and C_7 .

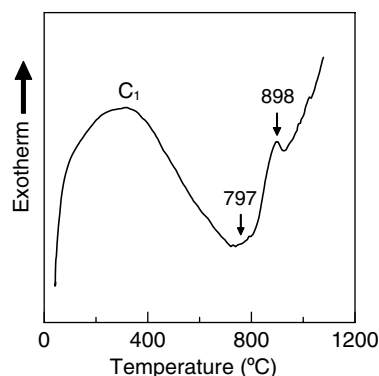


Fig. 1. DTA curve of C_1 at a heating rate of 5 °C/min.

Table 1
Stoichiometry of mixed alkali in $(1-x)Li_2O$ – xNa_2O – Al_2O_3 – $4SiO_2$ glasses (mol)

Sample	Li_2O	Na_2O	Al_2O_3	SiO_2
C_1	1	0	1	4
C_2	0.8	0.2	1	4
C_3	0.6	0.4	1	4
C_4	0.5	0.5	1	4
C_5	0.4	0.6	1	4
C_6	0.2	0.8	1	4
C_7	0	1	1	4

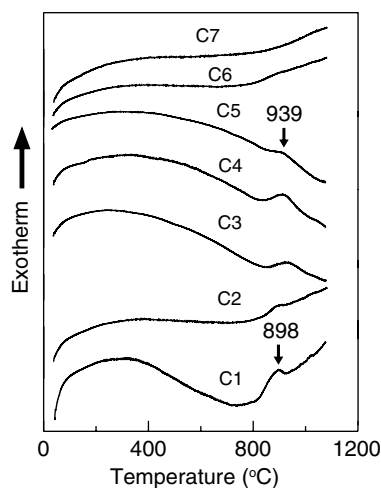


Fig. 2. DTA curves of $(1-x)\text{Li}_2\text{O}-x\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-4\text{SiO}_2$ glasses at a heating rate of $5^\circ\text{C}/\text{min}$.

The T_g , dilatometric softening temperature (M_g), and T_c of these glass samples are shown in Fig. 3. It indicates that the T_g increases when the Na_2O content increases from 0 to 1 mol. Subsequently, T_g decreases with Na_2O content increased from 0.6 to 0.8 mol. On the other hand, the T_g sharply increase when Na_2O content increases from 0.8 to 1 mol. Ray [21] has pointed out that the T_g , the temperature at which the network acquires mobility, changes from a rigid to a plastic structure. The T_g of these glasses is around $816 \pm 30^\circ\text{C}$, suggesting that the tetrahedra are the same mobile components of the glasses, except C_7 at 846°C [22].

From Fig. 3, it is also found that the M_g slightly decreases from 853.1 to 833.3°C when the Na_2O content increases from 0 to 0.6 mol. Furthermore, the M_g sharply decreases from 833.3 to 791.8°C when the Na_2O increases from 0.6 to 0.8 mol. When the Na_2O content increases from 0.8 to 1 mol, the M_g abruptly increases from 791.8 to 931.7°C .

Moreover, Fig. 3 also indicates that the T_c increases from 898 to 939°C when the Na_2O increases from 0 to

0.6 mol. This phenomenon has also been reported by Wang et al. [20] and Guides et al. [28]. On the other hand, the T_c of C_6 and C_7 does not appear when the Na_2O contents are 0.8 and 1 mol because of an abrupt change of the M_g .

3.2. Properties of the $(1-x)\text{Li}_2\text{O}-x\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-4\text{SiO}_2$ glasses

The dependence of the coefficient of thermal expansion (CTE) on the Na_2O content of the $(1-x)\text{Li}_2\text{O}-x\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-4\text{SiO}_2$ glasses are shown in Fig. 4. The CTE value increases from 6.65×10^{-6} to $10.07 \times 10^{-6}^\circ\text{C}^{-1}$ when the Na_2O content increases from 0 to 1 mol, but it is near $8.35 \times 10^{-6}^\circ\text{C}^{-1}$ when the Na_2O content varies from 0.2 to 0.6 mol. When the glass does not contain Na_2O (C_1) and contains Na_2O greater than 0.6 mol (C_6 and C_7), the CTE value sharply decreases and increases, respectively. This phenomenon indicates that the Na_2O addition increases the CTE value, but when the Na_2O content increase from 0.2 to 0.6 mol, i.e. the Li_2O content varies from 0.8 to 1 mol, the modification effect is delayed by Li_2O content with a slight variation in CTE. Furthermore, when the Na_2O content is greater than 10.00 mol%, the effect of Na_2O dominates over Li_2O , creating an abrupt increase in CTE.

The effect of Na_2O addition on the density of the $(1-x)\text{Li}_2\text{O}-x\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-4\text{SiO}_2$ glasses is shown in Fig. 5. It indicates that the density increases from 2.378 to 2.533 g cm^{-3} when the Na_2O content increases from 0 to 1 mol. The trend of the density variation is similar to the CTE, except the glass C_2 .

Fig. 6 shows the effect of mixed alkali on the electrical resistivity of the $(1-x)\text{Li}_2\text{O}-x\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-4\text{SiO}_2$ glasses. It is found that the electrical resistivity increases from 2.4×10^8 to $3.32 \times 10^9 \Omega \text{ m}$ when the Na_2O content increased from 0 to 0.4 mol. The electrical resistivity has a maximum value, namely $3.32 \times 10^9 \Omega \text{ m}$, at $\text{Na}_2\text{O} \cdot (\text{Na}_2\text{O} + \text{Li}_2\text{O})^{-1} = 0.4$ (the Na_2O content is 0.4 mol).

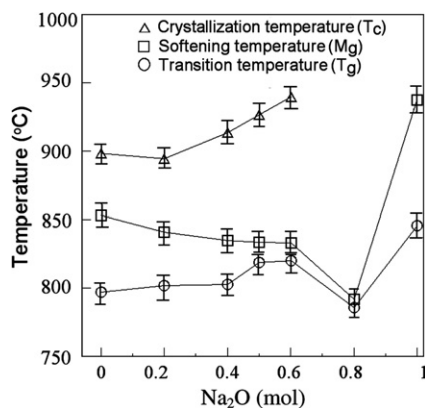


Fig. 3. Effect of Na content on T_g , M_g , T_c for $(1-x)\text{Li}_2\text{O}-x\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-4\text{SiO}_2$ glasses.

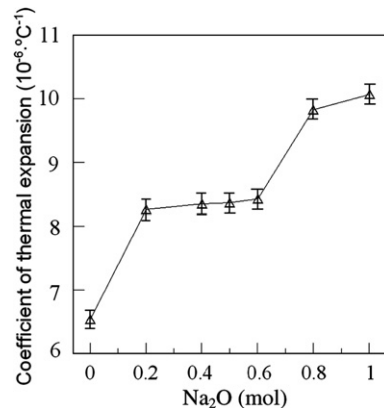


Fig. 4. Dependence of coefficient of thermal expansion on the Na_2O content of $(1-x)\text{Li}_2\text{O}-x\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-4\text{SiO}_2$ glasses.

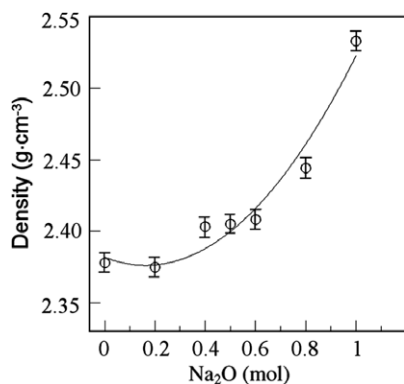


Fig. 5. Relation between density and Na₂O content for (1-x)Li₂O-xNa₂O-Al₂O₃-4SiO₂ glasses.

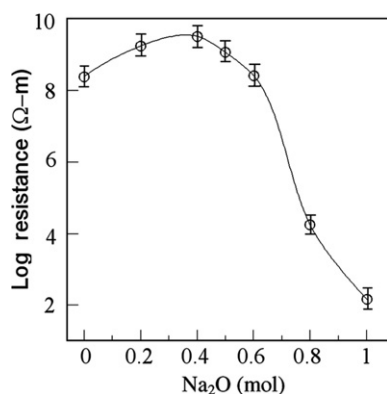


Fig. 6. Mixed alkali effect on electrical resistivity of (1-x)Li₂O-xNa₂O-Al₂O₃-4SiO₂ glasses.

3.3. Effect of Na₂O addition on the activation energy of crystallization

The T_c of the powder samples of the (1-x)Li₂O-xNa₂O-Al₂O₃-4SiO₂ glasses at various heating rates are shown in Fig. 7. It is found that the T_c shift to the higher temperature when the heating rate increases.

The activation energy of crystallization (E_c) can be calculated by Eq. (2) [23]

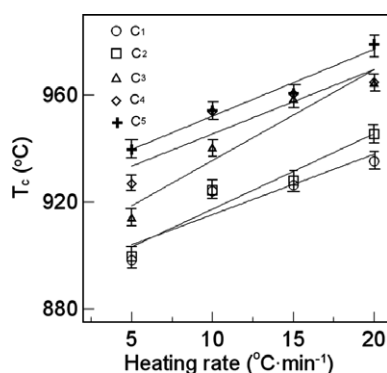


Fig. 7. Heating rate (β) vs. crystallization temperature (T_c) of the (1-x)Li₂O-xNa₂O-Al₂O₃-4SiO₂ glasses.

$$\ln \beta = -E_c(RT_p)^{-1} + \text{constant}, \quad (2)$$

where β is the heating rate ($^{\circ}\text{C min}^{-1}$), T_p is the exothermic peak temperature in the DTA curve, and R is the gas constant.

Fig. 8(a) shows the plot of $\ln \beta$ vs. $\frac{1}{T_c}$. The activation energy of crystallization of the (1-x)Li₂O-xNa₂O-Al₂O₃-4SiO₂ glasses can be obtained from the slope of each line. The dependence of the activation energy of the crystallization of glasses on the Na₂O content is shown in Fig. 8(b). It is found that the activation energy of crystallization decreases from 444 to 284 kJ mol⁻¹ when the Na₂O content increases from 0 to 0.4 mol and increases from 284 to 446 kJ mol⁻¹ when the Na₂O content increases from 0.4 to 0.6 mol which will be attributed to be solution effect of the system. As mentioned above, the Li₂O-Al₂O₃-4SiO₂ glass has liner non-bridging oxygen content in crystallization occurred with higher activation energy.

3.4. FTIR spectra

The FTIR spectra of the (1-x)Li₂O-xNa₂O-Al₂O₃-4SiO₂ glasses with five characteristic absorption regions are shown in Fig. 9. The most intense absorption band at 1054–1035 cm⁻¹ is caused by the antisymmetric stretching motion of SiO₄ tetrahedra containing bridging (i.e. Si-O-Si bond) and non-bridging (i.e. Si-O bonding) oxygens [24]. It is well known in fused silica that there is a continuous random network of SiO₄ tetrahedra and all the oxygens are doubly bonded, i.e. they are linked to two silicon atoms

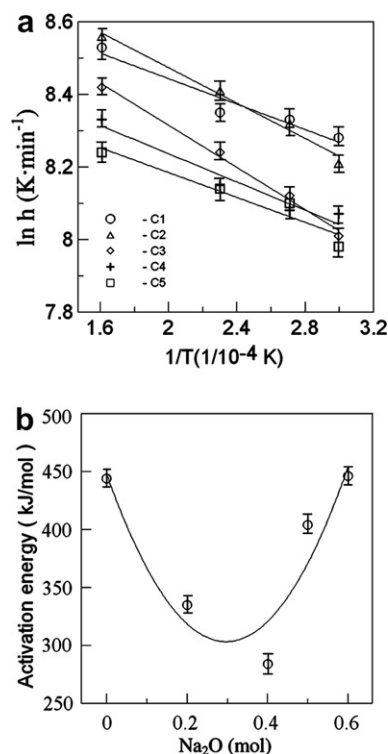


Fig. 8. (a) $\ln \beta$ vs. $\frac{1}{T_c}$ plot, and (b) activity energy of crystallization of (1-x)Li₂O-xNa₂O-Al₂O₃-4SiO₂ glasses.

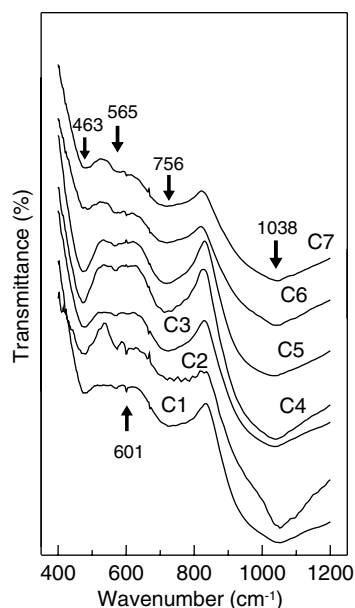


Fig. 9. FT-IR spectra of $(1-x)\text{Li}_2\text{O}-x\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-4\text{SiO}_2$ glasses.

and the tetrahedra do not share corners, but are linked through the apices. The absorption band lies at $763\text{--}713\text{ cm}^{-1}$ is due to symmetric $(\text{Si}, \text{Al})\text{--O--}(\text{Si}, \text{Al})$ stretching mode [24]. Partridge [6] has pointed out that the effect of smaller cations interacting stronger with the oxygen atoms involves the $(\text{Si}, \text{Al})\text{--O--}(\text{Si}, \text{Al})$ depending on the amount of both Al^{3+} in the network and alkali ions in the interstices for ensuring electroneutrality. When Al substitution for Si causes a shift-down of the stretching mode of the $\text{SiO}_4\text{--AlO}_4$ networks. The weak absorption band at $607\text{--}601\text{ cm}^{-1}$ is related to Si--O bending [25]. The absorption range at $573\text{--}565\text{ cm}^{-1}$ is assigned to the cavity occupancy by LiO_4 or NaO_4 tetrahedra [24]. The band at $486\text{--}463\text{ cm}^{-1}$ is due to the deformation mode of Al--O--Si or Al--O--Al bridges [24].

From the above result, it can be found that the change of glass structure is not expected to occur when the Na_2O is added to the $\text{Li}_2\text{O--Al}_2\text{O}_3\text{--}4\text{SiO}_2$ glass, forming the $(1-x)\text{Li}_2\text{O--}x\text{Na}_2\text{O--Al}_2\text{O}_3\text{--}4\text{SiO}_2$ glasses.

4. Discussion

The result of Fig. 3 causes the continuous disappearance of the high-viscosity phase, creating a rather abrupt change of the M_g occurred when the Na_2O content is greater than 0.6 mol. This phenomenon has also been found in the $\text{Li}_2\text{O}(\text{Na}_2\text{O})\text{--CaO--Al}_2\text{O}_3\text{--GeO}_2$ glasses [20]. Mazurin et al. [26,27] have pointed out that the effect of M_g on the phase-separated glasses and found that M_g is controlled by the higher viscosity phase as long as the phase is continuous.

Moreover, the phenomenon of Fig. 3 is due to that the Na_2O content greater than 10.00 mol% has a considerable content of the continuous high-viscosity phase which cre-

ates the difficulty of atom arrangement and maintains the amorphous state.

According to the result of Fig. 6, Tomozawa [29] has pointed out that the alkali ion mobility in oxide glasses decreases drastically when a second alkali ion is introduced, causing by an increase in activation energy. Moreover, the maximum magnitude of the electrical resistivity upon the second alkali is accounted for by compositional dependence of the alkali ion mobility as reported by Day [8] and Varsheneya [30]. On the other hand, when the alkali ions transport in the $(1-x)\text{Li}_2\text{O--}x\text{Na}_2\text{O--Al}_2\text{O}_3\text{--}4\text{SiO}_2$ glasses, it is expected that Na^+ ion occupies an Li^+ ion vacancy and Li^+ ion occupies a Na^+ ion vacancy. Since the ionic motion is faster than the rate of structural relaxation [31], these Na^+ and Li^+ ions would be occupying ‘wrong sites’ [29]. Na^+ and Li^+ ions in ‘wrong sites’ are constantly created and annihilated by the motion of the alkali ions and the stress relaxation [29]. Hence, the electrical resistivity attains a maximum value.

Furthermore, the electrical resistivity rapidly decreases from 3.32×10^9 to $1.47 \times 10^2 \Omega\text{ m}$ when the Na_2O addition increases from 0.4 to 1 mol. The lower electrical resistivity is caused if the alkali ions in ‘wrong sites’ have occurred and not annihilated when the Na_2O content increases. Since the electrical resistivity of the mixed alkali glasses is attributed to a change in both distribution and size of these compositionally different regions, which are dependent on the total alkali content and different alkali ions present. The activation energy barrier for alkali ion mobility is influenced by the media surrounding the alkali. When the Na_2O addition increases to 0.8 mol the different alkali ions present decreases and introduces the activation energy barrier for alkali ion mobility, which creates decreased electrical resistivity. Moreover, the electrical resistivity of the $\text{Na}_2\text{O--Al}_2\text{O}_3\text{--}4\text{SiO}_2$ glass (C_7) is smaller than the $\text{Li}_2\text{O--Al}_2\text{O}_3\text{--}4\text{SiO}_2$ glass (C_1). This result suggests that the non-bridging oxygen content in the C_7 is greater in C_1 , hence C_7 has the lowest electrical resistivity in the $(1-x)\text{Li}_2\text{O--}x\text{Na}_2\text{O--Al}_2\text{O}_3\text{--}4\text{SiO}_2$ glasses.

5. Conclusions

The properties and structure of the $(1-x)\text{Li}_2\text{O--}x\text{Na}_2\text{O--Al}_2\text{O}_3\text{--}4\text{SiO}_2$ glasses have been investigated. The results are summarized as follows:

1. In DTA curves, the crystallization temperature increases from 898 to 939°C when the Na_2O content increases from 0 to 0.6 mol. Moreover, when the Na_2O content is greater than 0.6 mol, the exothermic peak of crystallization in the DTA curve is not apparent.
2. The glass transition temperature (T_g) increases when the Na_2O content increases from 0 to 0.6 mol and subsequently decreases when the Na_2O content increases from 0.6 to 0.8 mol. If the Na_2O content increases from 0.8 to 1 mol, the T_g sharply increase. The variation trend of dilatometric softening temperature (M_g) is similar to T_g .

3. The coefficient of thermal expansion and density of the $(1-x)\text{Li}_2\text{O}-x\text{Na}_2\text{O}-\text{Al}_2\text{O}_3-4\text{SiO}_2$ glasses increase from 6.54×10^{-6} to $10.1 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$ and 2.378 to 2.533 g cm^{-3} when the Na_2O content increases from 0 to 1 mol, respectively.
4. The electrical resistivity has a maximum value, namely, $3.32 \times 10^9 \text{ } \Omega \text{ m}$, at $\text{Na}_2\text{O} (\text{Na}_2\text{O} + \text{Li}_2\text{O})^{-1} = 0.4$ (i.e. Na_2O content is 0.4 mol).
5. The activation energy of crystallization decreases from 444 to 284 kJ mol^{-1} when the Na_2O content increases from 0 to 0.4 mol, and increases from 284 to 446 kJ mol^{-1} when the Na_2O content increases from 0.4 to 0.6 mol, which will be attributed to the solution effect of the system.
6. The FTIR spectra show that when Na_2O is added, the significant change of structure of these glasses does not occur.

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